Bogdan Alexandru Stoica

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Research Interests

My research interests focus on software reliability, security, and efficiency with an emphasis on automated debugging at scale. I have experience building tools to help developers better reason about their code and diagnose bugs faster. I am interested in program analysis, efficient code instrumentation, emerging hardware and OS support for program monitoring, compiler optimization techniques and software performance analysis.

Education

University of Chicago — Chicago, IL, United States

Ph.D. Student in Computer Science (Area: Systems and Programming Languages) — September 2019 – *present*

Advisor: Prof. Shan Lu

M.Sc. in Computer Science (Area: Systems and Programming Languages) — September 2019 – June 2022

Thesis: Exposing Memory Ordering Bugs Efficiently with Active Delay Injection
Advisor: Prof. Shan Lu
Mentors: Dr. Suman Nath and Dr. Madan Musuvathi (Microsoft Research)

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland

Graduate Coursework (Area: Systems and Programming Languages) — September 2014 – August 2015

Mentors: Prof. Bernard M. E. Moret (EPFL), Prof. Viktor Kunčak (EPFL) and Prof. Vikram S. Adve (UIUC)

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland

M.Sc. in Communication Systems (Area: Information Security) — September 2011 – January 2014

Thesis: Robust Web Content Evaluation
Advisor: Prof. Karl Aberer

University of Bucharest (UB) — Bucharest, Romania

B.Sc. in Computer Science (Area: Algorithms and Software Security) — October 2008 – June 2011

Thesis: On Intrusion Detection Systems
Advisor: Prof. Adrian Atanasiu

Peer-Reviewed Publications

[C1] **Waffle: Exposing Memory Ordering Bugs Efficiently with Active Delay Injection.** Bogdan Alexandru Stoica, Shan Lu, Madan Musuvathi, and Suman Nath. *In Proceedings of the 18th ACM SIGOPS European Conference on Computer Systems, Rome, Italy, 2023 (to appear).* — EuroSys '23

Awards and Honors

Eckhardt Graduate Fellowship (University of Chicago)	2019 – 2024
Teaching Assistant Award for Outstanding Service (EPFL)	2016 – 2017
Excellence Fellowship for Master's Studies (EPFL)	2011 – 2013

Excellence Scholarship for Bachelor's Studies (UB) 2008 – 2011
Bronze Medal, Baltic Olympiad in Informatics (Tuymaada), Russian Federation 2007

Employment

University of Chicago Chicago, IL, United States
Research Assistant September 2019 – present
Advisor: Prof. Shan Lu

I develop automated fault diagnosis techniques for large-scale distributed software systems written in C, C++, C#, and Java. I build tools that combine efficient code instrumentation, execution monitoring, and program analysis to isolate and expose correctness and performance bugs.

Meta Seattle, WA, United States
Research Intern (Systems & Infrastructure, Profiling) June 2022 – October 2022
Mentors: Nathan Slingerland and Jacie Fan

I investigated inefficient C++ code patterns that surface at scale, and developed a rules-based engine to identify them by analyzing billions of low-level execution profiles collected across multiple Cloud services. This tool is currently being productized inside the compnay to enable developers pinpoint difficult-to-detect performance bottlenecks fleet-wide.

Microsoft Research Redmond, WA, United States
Research Collaborator January 2020 – June 2020
Mentors: Dr. Suman Nath and Dr. Madan Musuvathi

I explored various bug diagnosis techniques that can easily integrate at scale with existing software testing frameworks. In particular, I investigated several fault injection strategies to help expose difficult-to-reproduce order violations in C# distributed systems. This exploratory work is a precursor of Waffle, a push-button fault injection tool that helps developers trigger memory ordering concurency bugs [C1].

Microsoft Research Redmond, WA, United States
Research Intern July 2018 – September 2018
Mentor: Dr. Xinyang Ge

I designed and implemented a C/C++ tool for tracking heap memory allocation/deallocation requests to aid program state recovery during offline execution replay. The goal was to extend the current reverse debugging engine inside the Windows Debugger, increasing program state recovery rate and replaying longer execution traces.

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Research Assistant September 2015 – December 2018
Mentors: Prof. Vikram S. Adve (UIUC), Prof. Bernard M. E. Moret (EPFL), and Dr. Swarup K. Sahoo (CMU)

I developed techniques to help developers better reason about their code and improve software reliability. I prototyped several C/C++ tools for scalable software analysis using program analysis, efficient code instrumentation, and emerging hardware support for execution tracing.

Microsoft Prague, Czech Republic
Software Development Engineer (Automated Testing Infrastructure) February 2014 – August 2014
Mentor: Travis Merkel

I worked on developing automated testing frameworks (C/C++, Python) for the Skype tool chain, including for web plug-ins. I used the testing framework to design performance testbeds which identified critical memory leaks and vulnerabilities.

Bitdefender Labs Bucharest, Romania

Software Development Engineering Intern (R&D) June 2012 – September 2012
Mentor: Teodor Stoenescu

I developed a stand-alone, secondary SSL certificate validation tool for the Bitdefender Anti-virus suite (C/C++). Parts of my code were integrated in the 2013 release.

Bitdefender Labs Bucharest, Romania
Software Development Engineer (R&D) May 2010 – August 2011
Mentors: Mihai Chiriac and Teodor Stoenescu

I developed several modules of a new Anti-virus suite (C/C++) for virtual environments. I focused on optimizing network traffic processing and implemented a multi-layer cache which increased scanning throughput by 50%.

Invited Talks

Hardware-Assisted Program Analysis
University of Illinois at Urbana-Champaign, Champaign, IL, USA Sept. 2018

Exploring Hardware Data Logging on Modern CPUs
Microsoft Research, Redmond, WA, USA Sept. 2018

Modern Hardware and OS Support for Efficient Execution Tracing
University of Zurich, Zurich, Switzerland Dec. 2017

Professional Service

Symposium on Operating Systems Design and Implementation (OSDI), Artifact Evaluation Committee 2022
European Conference on Computer Systems (EuroSys), Artifact Evaluation Committee 2022
Intl. Conf. on Arch. Support for Prog. Lang. and Operating Systems (ASPLOS), Artifact Evaluation Committee 2022
Symposium on Operating Systems Principles (SOSP), Artifact Evaluation Committee 2021
Intl. Conf. on Programing Languages Design and Implementation (PLDI), Artifact Evaluation Committee 2019
Symposium on Principles and Practice of Parallel Programming (PPoPP), Artifact Evaluation Committee 2018, 2019

Teaching Experience

MPCS-52030: Operating Systems (UChicago, graduate), teaching assistant 2020
CAPP-30122: Computer Science with Applications II (UChicago, graduate), teaching assistant 2020
MPCS-55001: Algorithms (UChicago, graduate), teaching assistant 2019

CS-173: Digital Systems Design (EPFL, undergraduate), lead teaching assistant 2018
CS-207: Systems Oriented Programming (EPFL, undergraduate), lead teaching assistant 2015, 2017
CS-250: Algorithms (EPFL, undergraduate), teaching assistant 2015, 2016, 2017
CS-450: Advanced Algorithms (EPFL, graduate), teaching assistant 2013, 2014, 2016
CS-150: Discrete Structures (EPFL, undergraduate), teaching assistant 2013

Research Mentoring

Liam Fox (2nd year undergraduate) June 2017 – August 2017

Liam and I worked on a tool that computes imprecise dynamic program slices on x86 binaries starting from partial data-flow trace information as part of the *Statistical Program slicing* Project.

Siddhant Garg (3rd year undergraduate) May 2017 – July 2017

Siddhant and I worked on optimizing the decoding library for Intel Processor Trace byproducts relying on clever static program analysis (LLVM).

Bobbie Lynn Eicher (2nd year Master's) June 2016 – August 2016

Bobbie and I worked on an early dynamic program slicing prototype using DynamoRIO as a dynamic instrumentation framework.

Kishalay Raj (3rd year undergraduate) May 2016 – July 2016

Kishalay and I worked on static program slicing (LLVM) and set the foundation for a more efficient program slicing technique to be used on large-scale code bases (e.g., 100K+ lines of code).

Van Quoc Dung (1st year Master's) June 2015 – August 2015

Van and I analyzed execution patterns in production runs using static instrumentation (LLVM) and software path profiles to characterize deviant program behavior.

Languages

English – fluent
French – intermediate
German – beginner
Romanian – native

References

Available upon request.